Chain returns and lower bounds for superpermutations

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Abstract

A short superpermutation must make extensive use of large overlaps between successive permutations. These overlaps organize the permutations into runs, and earlier lower bounds control how efficiently such runs can be joined. We strengthen this part of the argument. The patterns that attain the earlier estimates cannot keep repeating: their action on the symbols eventually brings them back to a rotation class already visited. In one case, two repetitions already have an entire run in common. Counting the necessary departures from these patterns gives an explicit lower bound valid for every alphabet size at least eight. It improves the preceding general formula at every size at least thirteen not divisible by three. The leading correction above $K! + (K-1)! + (K-2)!$ remains asymptotic to $2(K-4)!$. The exact formulas and their asymptotic comparisons have formal proofs in Lean.

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1 Introduction

A *superpermutation* on K symbols is a word containing each of the $K!$ permutations as a consecutive subword. For example, 123121321 contains all six permutations on three symbols. Write $S(K)$ for the length of a shortest superpermutation and put

$$F_3(K) = K! + (K-1)! + (K-2)!.$$

The lower bounds studied here exceed $F_3(K)$ by a quantity asymptotic to $2(K-4)!$. We improve the smaller terms by examining when the local counting estimates can be tight. The resulting inequalities hold at every alphabet size in their stated range.

The difficulty is that a pattern may be efficient for a few steps without remaining efficient for very long. One can see the simplest instance directly in a word. Appending the first symbol of a permutation produces its next cyclic rotation, at a cost of only one letter. Repeating this operation is inexpensive, but after K steps it returns to the original permutation. A covering word must eventually do something else.

The same phenomenon occurs further into the lower-bound argument. After grouping rotations together, and then grouping those classes into runs, the relevant objects are chains of almost full runs. The earlier proof bounds the length of each chain and restricts which lengths may follow one another. Its best average can nevertheless be attained by a repeating sequence of chain lengths. Such a sequence does not determine fresh permutations indefinitely. The corresponding relabeling of the symbols has finite order, and when it returns, so does a previously visited class.

This gives a way to improve the average. Each interruption of the repeating pattern loses some of the length allowed by the old estimate. If there is a uniform bound on the distance between interruptions, these losses accumulate. There are two parts to making this observation useful: finding a sufficiently short return, and proving that all other patterns really do lose enough to pay for the interruptions. Most of the work in this paper concerns the second part.

Our explicit bound is stated in Theorem 6.2. It applies for every $K \geq 8$, and incorporates the earlier refinement for K divisible by three. In each of the other two residue classes it strictly improves the preceding general formula for every $K \geq 13$. At $K = 13$ and $K = 14$, for example, the increases are 12 and 58, respectively. The improvements hold uniformly in the alphabet size; no separate search is used to prove an instance.

We use the geometric and forest lemmas of the lower-bound framework retained in the accompanying proof library [1]. These lemmas reduce an arbitrary word to auxiliary paths of complete rotation classes. The new estimates in this paper concern those paths. The next section explains why counting deficits in such paths gives a word-length bound. We then describe the chains that make the count difficult. A fourteen-symbol example introduces the return argument before we prove it in general. The explicit numerical formula is given after these arguments, when its parameters have a combinatorial meaning. The coordinate calculations and the endpoint count for the forest reduction are given in the appendices.

2 Overlaps, runs, and deficits

To understand the quantities in the lower bound, we first describe how the cheapest overlaps fit together. The *weight* of a step from a permutation u to a different permutation v is the number of letters that must be appended to a word ending in u to make it end in v. Thus the weight is K minus the length of the longest suffix of u that is a prefix of v. Ordering one chosen occurrence of each permutation in a covering word produces a Hamiltonian path whose total weight is at most the word's length minus K. Conversely, appending the required letters along such a path produces a covering word. Hence $S(K)$ is K plus the minimum weight of a Hamiltonian path in this overlap graph.

A step has weight one precisely when it passes to the next cyclic rotation. We call the K rotations of a permutation a *rotation class*. Traversing all of them in cyclic order gives a *complete block*. The class of 1234, for instance, consists of $1234, 2341, 3412, 4123$. Within this block, each of the three steps costs one. Repeating the next step would revisit 1234.

Our arguments take place in paths of complete blocks whose rotation classes are pairwise distinct. The minimum-path and forest reduction produces auxiliary paths of this kind from an arbitrary superpermutation [1]. Its exact form is recalled in Appendix D. A superpermutation as written need not itself traverse whole classes at once. We shall return to this reduction after seeing what can be counted in one of its paths.

It is convenient to put $n = K - 1$ and use the alphabet $\{0, \ldots, n\}$. Suppose a block is entered at $(x_1, \ldots, x_n, s)$. After traversing the block, the exit is $(s, x_1, \ldots, x_n)$. A weight-two step to a different rotation class must enter at $(x_2, \ldots, x_n, x_1, s)$. The other ordering of the two remaining symbols lies in the class just visited. Thus the candidate in a different class rotates the first n symbols and leaves the last symbol fixed. It may still be excluded if that class has appeared earlier in the path. Successive weight-two steps therefore arrange the blocks themselves around a cycle of length n.

Figure 1 shows this at $K = 5$. Each row is one block, and the four rows are joined by weight-two steps. Continuing would bring us back to the first row, so a class-distinct path can use at most four of these blocks consecutively.

12340	23401	34012	40123	01234
23410	34102	41023	10234	02341
34120	41203	12034	20341	03412
41230	12304	23041	30412	04123

Figure 1: A full run on five symbols. Read each row from left to right, then pass to the next row. Steps within a row have weight one; steps between rows have weight two. The same weight-two step from the last row returns to the first.

A maximal succession of blocks joined at weight two is a *run*. If it contains L blocks, its *deficit* is $n - L$. A full run has deficit zero. For example, stopping Figure 1 after the third row gives a run of deficit one. A path with r runs and total deficit D thus contains $nr - D$ blocks.

This identity explains the importance of deficits. They measure how much of the capacity of the runs goes unused. If the total number of blocks is fixed, a larger deficit forces more runs, and passing from one run to the next costs an extra letter.

For a precise illustration, suppose a single path visits all $T = n!$ rotation classes and every boundary between runs has weight three. There are T blocks, $T - r$ weight-two boundaries, and $r - 1$ weight-three boundaries. Its word length is therefore

$$K + (K-1)T + 2(T-r) + 3(r-1) = K + (K+1)T - 3 + r. \tag{1}$$

Ignoring deficits would give at least $T/n = (n-1)!$ runs. Substituting this into (1) gives $F_3(K)+K-3$. Thus the familiar factorial terms already appear in this simple count; the extra contribution comes from proving that runs leave some capacity unused.

The quantity we need to bound is the number of runs. Since $T = nr - D$, an estimate $D \geq \delta r - \beta$ gives $T \leq (n - \delta)r + \beta$. In other words, apart from the endpoint allowance β, each run covers at most $n - \delta$ classes on average. The denominator $n - \delta$ in the eventual lower bound has this meaning.

The forest reduction extends this accounting to arbitrary words. It handles paths that break into components and boundaries of weight greater than three, while keeping the same consequence of a deficit estimate. The estimate must hold on every consecutive segment of runs, because the proof applies it after cutting paths and adding certain entrance blocks. This is the scope of the local arguments below.

3 The chains that resist the count

For the remainder of the paper, assume $n \geq 7$. We now ask how a path can have many runs while keeping its deficit small. Restrict attention to consecutive runs joined at weight three; call such a path a *genuine light segment*. Its blocks still belong to distinct rotation classes. Record the segment by its sequence of run deficits. For example, an entry 2 represents a run missing two blocks, while an entry 0 represents a full run.

The local overlap restrictions prevent us from inserting arbitrarily many zeros between small positive deficits. Put $a = n - 2$. A string of full runs has length at most a, unless the whole segment consists of full runs, in which case its length is at most $a + 1$. The pattern $(0, 1, 0)$ is impossible: a run missing one block cannot have full runs on both sides. More generally, a factor $(0, 1, d)$ has $d \geq 2$. Finally, if a gap of a full runs separates two positive deficits, those adjacent deficits sum to at least $a + 2$. These are the earlier local run restrictions [1].

Thus the first small positive deficit that can sit alone between full runs is 2. Two such entries can be separated by at most $a - 1$ zeros. The pattern that is most troublesome for the deficit count

consists of many singleton twos separated by these long gaps. Each two contributes the same deficit, while the zeros increase the number of runs being covered by the estimate.

This leads to the chains used in the proof. In a factor starting and ending in zero, a positive cluster is *regular* if it is the singleton 2. Join consecutive regular clusters across gaps of $a-1$ zeros to form a maximal *chain*. Its length is its number of regular clusters. A gap of $a-2$ zeros joins consecutive chains into a *path of chains*; shorter gaps and other positive clusters cut the path. At $a=5$, for example, the factor

$$0,\ \underbrace{2,0,0,0,0,2}_{\text{chain of length 2}},\ \underbrace{0,0,0}_{\text{shortened gap}},\ \underbrace{2}_{\text{chain of length 1}},\ 0$$

has two chains. These definitions retain the original adjacencies: cutting out a nonregular cluster does not join its two neighboring gaps.

There are two ways to improve the count. One can limit the length of a single chain, or show that several long chains cannot follow one another too efficiently. The earlier proof already does both. We shall strengthen the second estimate. The form we need is

$$\ell_1 + \cdots + \ell_q \le p + \frac{t}{2}(q-1) \tag{2}$$

for every path of q chains. Here p is an endpoint allowance, while $t/2$ is the limiting average chain length permitted by the estimate. Write $\mathrm{CP}(p,t)$ if (2) holds in every zero-ended factor of every genuine light segment.

Each boundary between chains has one fewer full run than a link within a chain. For a fixed number of regular clusters, more such boundaries therefore allow fewer zero entries without changing the total deficit. This is why shortening the permitted average chain length strengthens the deficit estimate. The earlier scalar argument makes this precise: all the nonregular clusters and shorter gaps already supply enough deficit to cover their part of the count. We recall that argument in Appendix C.

3.1 Following a chain also relabels the symbols

The deficit sequence alone does not record which classes have been visited. To recover some of this information, follow a chain in actual permutations. Denote the two cells by

$$C = (2, 0^{a-1}), \qquad B = (2, 0^{a-2}),$$

where an exponent means repetition. A chain of length ℓ followed by a shortened gap has pattern $T_\ell = C^{\ell-1}B$. The final B consists of the chain's last deficit-two run and its outgoing shortened gap; this description is used only when that gap is present.

The cells are particularly useful because their internal steps are forced. To see what this means, relabel a current exit as $(0,1,\dots,n)$. A weight-three step leads to $(3,\dots,n,i,j,k)$, where (i,j,k) is one of the six orders of $(0,1,2)$. The local cell calculation forces the order 210. After a completed C, the same choice is forced if the next run has at least two blocks; after B, it is forced if the next run has at least three. The alternatives collide with classes in the cell just traversed [1].

Consequently, once the first entry is fixed, each cell carries us to a specified final exit. Relabeling that exit back to $(0,1,\dots,n)$ puts the next cell into the same coordinates. Call this relabeling the cell's *normalization*. It acts on the visited classes as well as on the exit. This last point is essential: resetting the coordinates does not erase the earlier part of the path.

In one-line notation on $0, \ldots, n$, the two normalizations are

$$g = (0, 5, 4, 6, \ldots, n, 2, 3, 1), \qquad \rho = (0, 1, 3, 4, \ldots, n, 2), \qquad N(C) = g, \quad N(B) = \rho g. \tag{3}$$

Products act rightmost first, so the normalization of T_ℓ is $h_\ell = \rho g^\ell$. The maps compose in reverse chronological order. If a succession of completed chains has normalization equal to the identity, its actual following short-run start repeats the first start class. That is impossible in a genuine segment.

Let p_0 be the order of g. The earlier return calculation gives

n	p_0
$3m$	$2m$
$3m+1$	$m(2m+1)$
$3m+2$	$(m+1)(2m+1)$.

For example, at $n = 7$ the nonzero cycles of g are $(1, 5, 2, 4, 7)$ and $(3, 6)$, giving order 10. A single chain has length at most p_0. If there is more than one chain in the path, the earlier endpoint collisions reduce every chain's maximum to $b = p_0 - 1$ [1].

Knowing these maxima does not yet give the best average. A very long chain can force its successor to be short. The next two sections examine the patterns that attain the resulting averages, beginning with a case where there is just one efficient pair.

3.2 The inherited restrictions on consecutive chains

We collect the local geometric facts used in the counts below. The parameters p_0 in this statement are the orders just displayed. These are restrictions on actual occupied classes, not on arbitrary integer sequences with the same deficit sums.

Lemma 3.1 (Local chain restrictions [1])**.** *Every chain has length at most p_0. In a path of at least two chains, every length is at most $b = p_0 - 1$. The following additional restrictions hold whenever the indicated consecutive chains occur.*

For $n = 3m + 1$ and $m \geq 3$, in two consecutive chains, if the first is longer than m, then the second has length at most m. That successor has length at most $m - 1$ if it is not the last chain of the path.

For $n = 3m + 2$ and $m \geq 2$, put $v = 2m(m + 1)$. In two consecutive chains, if the first is longer than v, then the second has length at most $v - 1$. If the first chain has length at least $v + 2$ and its successor is nonterminal, the successor has length at most $v - 2$.

For $n = 3m$ and $m \geq 3$, four consecutive chains cannot have their middle two lengths equal to b and both outer lengths at least two. For $n = 7$, the chain inequality $\mathrm{CP}(10, 10)$ *holds.*

The cell forcing rules and these chain restrictions are the inherited geometric inputs to the present argument. Their proofs identify a class in common between the relevant cell paths. Further occupied classes can exclude more paths, but cannot undo such a collision. This is why the same restrictions apply inside every factor to which the forest count will be applied.

4 Efficient pairs cannot keep repeating

We begin with the fourteen-symbol case, because the improvement can be seen before introducing its general parameters. Here $n = 13$. A chain has length at most 36, and in a path with two or more chains its length is at most 35. The earlier local collision restrictions say more: a chain longer

than four forces its successor to have length at most four, or at most three if that successor is not the last chain in the path.

Pair each nonterminal chain longer than four with its successor. The pairs are disjoint, because the successor is too short to begin another pair. All remaining chains, except possibly the last, have length at most four. An interior pair has total length at most 38, attained only by $(35, 3)$. Thus the earlier count allows an average of 19 regular clusters per chain. The length count by itself would permit $(35, 3), (35, 3), \ldots$ indefinitely.

Now follow what such a pair does to the symbols. Its normalization has nonzero cycles

$$(1\ 11\ 7\ 3\ 9), \qquad (2\ 10\ 5\ 12\ 6), \qquad (4\ 13\ 8).$$

After fifteen pairs the relabeling is the identity. If another chain then starts, it starts in the same rotation class as the first one. Consequently fifteen consecutive interior pairs cannot all be $(35, 3)$. At least one must have smaller total length, or the pairing must be interrupted by a singleton. Either event loses at least one regular cluster from the old allowance.

There are at most two chains per pair or singleton. Thus one lost unit per fifteen such pieces predicts a reduction of $1/30$ in the mean chain length. Equivalently, the parameter $t = 38$ should decrease by $1/15$. This is the source of the new value $569/15$. The argument below proves this prediction while allowing for the final chain, whose successor need not exist.

4.1 Counting the interruptions

Call a singleton or a chosen pair a *tile*. If the old mean chain length is μ, a tile containing k chains is allowed total length μk. Its *loss* is the difference between this allowance and its actual total length. This loss counts regular clusters; the run deficits introduced earlier count missing blocks.

The count is simple once the efficient tiles have been identified. Mark every tile with positive loss. These marks divide the zero-loss tiles into consecutive strings. A bound on the lengths of those strings then forces enough marks. The only endpoint issue is that the last tile may attain a larger length, and a return after it supplies no contradiction unless a further chain actually starts. We therefore set that tile aside.

Lemma 4.1. *Suppose a path of $q \geq 2$ chains is partitioned into tiles of one or two chains, and a tile with k chains and total length L has loss $\mu k - L$. Suppose that before the last tile every loss is nonnegative, every positive loss is at least $\varepsilon > 0$, and every consecutive string of zero-loss tiles has length at most $R - 1$, where $R \geq 1$. If the last tile has loss at least $-c$, where $c \geq 0$, then*

$$\sum_{i=1}^{q} \ell_i \leq \left(\mu - \frac{\varepsilon}{2R}\right) q + c + \varepsilon.$$

Proof. Let B_+ be the number of marked tiles and P their total loss, excluding the last tile. Then $B_+ \leq P/\varepsilon$. There are at most $B_+ + 1$ intervening strings, each with at most $R - 1$ zero-loss tiles. Restoring the last tile, the total number of tiles is at most $R(B_+ + 1)$. Since each tile has at most two chains,

$$q \leq 2R(P/\varepsilon + 1), \qquad P \geq \frac{\varepsilon q}{2R} - \varepsilon.$$

The total chain length is at most $\mu q - P + c$, giving the result. □

4.2 The case $n = 3m + 1$

The fourteen-symbol example is typical of this residue class. Assume $m \geq 3$, put $p_0 = m(2m + 1)$ and $b = p_0 - 1$, and pair every nonterminal chain longer than m with its successor. Lemma 3.1 bounds that successor by m, and by $m - 1$ when it is nonterminal. Hence an interior pair has total length at most $t_0 = p_0 + m - 2$, with equality only for $(b, m - 1)$.

Take $\mu = t_0/2$. An interior singleton has length at most m, so its loss is at least $\mu - m \geq 1$. All positive interior losses are integers, since $t_0 = 2m^2 + 2m - 2$ is even. Thus a tile either is the efficient pair $(b, m - 1)$ or loses at least one unit.

The normalization of that pair is

$$M_1 = h_{m-1}h_b = \rho g^{m-1}\rho g^{-1}. \tag{4}$$

Let R_1 be its order. As in the example, R_1 consecutive efficient interior tiles would return to the first start class. The next start is forced because a deficit-two run has $n - 2 \geq 8$ blocks, above the three-block threshold after B.

Merely knowing that M_1 has finite order would give a weak quantitative estimate. Here its cycles can be calculated explicitly. Their lengths are listed in Table 1, and the calculation in Appendix A proves $R_1 \leq n^2/4$. The example $n = 13$ has cycle lengths $5, 5, 3$, so $R_1 = 15$.

It remains to account for the last tile. A final singleton can have length b, giving loss $-(p_0 - m)/2$. A final pair can have total length $b + m = t_0 + 1$, giving loss -1. Thus the single allowance $c = (p_0 - m)/2$ covers both possibilities. Apply Lemma 4.1 with $\varepsilon = 1$. Since $p_0 - \mu = c + 1$, it yields

$$\sum_i \ell_i \leq \left(\mu - \frac{1}{2R_1}\right) q + c + 1 \leq p_0 + \frac{1}{2}\left(t_0 - \frac{1}{R_1}\right)(q - 1).$$

The case of a single chain already has the stronger bound p_0. We have therefore proved

$$\mathrm{CP}\left(p_0, t_0 - \frac{1}{R_1}\right). \tag{5}$$

At fourteen symbols, the final-tile allowance is 16. It remains an endpoint allowance; it does not cancel the losses that accumulate along a long path.

4.3 When the interiors collide first

For $n = 3m + 2$, following the first class of each tile misses an earlier collision. It is useful to recall the full run in Figure 1. Delete the symbol zero from each of its four rotation classes. All four give the same cyclic order 1234; they differ only in the gap into which zero was inserted. In general, a full run with distinguished symbol zero visits all n such insertions into one cyclic order of the other symbols. This family of classes is called a *two-loop*.

Consequently two full runs with the same nonzero cyclic order repeat every class, even if the starts of the larger pieces containing them are different. This is what happens in the efficient pattern for the present residue. Put $p_0 = (m + 1)(2m + 1)$, $v = 2m(m + 1)$, $b = p_0 - 1$, and $c = v - 2$.

Lemma 4.2. *For $m \geq 2$, the pattern $T_bT_cT_bT_c$ cannot occur in a genuine light segment.*

The repeated runs lie inside the two shorter chains. To locate them, look at the C cells of each $T_c = C^{c-1}B$. In the first copy choose the cell at distance m from the end of this C portion; in the second choose the cell at distance m from its beginning, as in Figure 2. The last full run of the first chosen cell and the first full run of the second chosen cell have the same nonzero cyclic order.

first pair	T_b	$C^{c-2-m}\boxed{C}C^mB$	last full run in the box
second pair	T_b	$C^m\boxed{C}C^{c-2-m}B$	first full run in the box

Figure 2: The two full runs that coincide. The boxes are at equal depths from opposite ends of the C portions of the shorter chains. Their cyclic orders are compared in the coordinates of the whole path.

Appendix B checks this equality by following the relabelings. Both runs lie strictly before their chains' final B cells, so the collision occurs within the displayed pattern.

We can now use the same loss count. Lemma 3.1 bounds the successor of a chain longer than v by $v-1$. If the first length is at least $v+2$ and the successor is nonterminal, the stronger bound is $v-2$. Pair every nonterminal chain longer than v with its successor, and put $t_0 = b+v-2 = 4m^2+5m-2$ and $\mu = t_0/2$.

For $m > 2$, the only interior pair attaining t_0 is $(b, v-2)$. A pair beginning with $v+1$ loses at least $m-2$, and every other positive pair loss is an integer. A singleton loses at least $(m-2)/2$. Thus every positive interior loss is at least $\varepsilon = \min\{1, (m-2)/2\}$. Lemma 4.2 excludes two successive efficient tiles, so take $R = 2$.

A final singleton has loss at least $-(m+2)/2$, and a final pair has loss at least -1. Lemma 4.1, together with $p_0 - \mu = (m+2)/2 + 1$ and $\varepsilon \le 1$, gives

$$\mathrm{CP}(p_0, t_0 - \varepsilon/2), \qquad m > 2. \tag{6}$$

The decrement is $1/4$ at $m = 3$ and $1/2$ thereafter.

At $m = 2$ there are three efficient tiles: (12), $(13, 11)$ and $(14, 10)$. Their normalizations lie in a group of order 12, so twelve consecutive efficient interior tiles would have two equal prefix normalizations and hence a repeated start class. Positive losses are integers and the final allowance is two. The same lemma gives $\mathrm{CP}(15, 287/12)$. The small group calculation is also included in Appendix B.

All these collisions occur within the path under consideration. Thus the counts apply in every zero-ended factor, as required by CP; no completion of such a factor to a superpermutation is assumed.

5 When the efficient pattern alternates

Assume $n = 3m$ with $m \ge 3$. This case has a different efficient pattern. Here the period is $p_0 = 2m$, so every chain in a nontrivial path has length at most $b = p_0 - 1$. Lemma 3.1 forbids four consecutive chains whose middle two both have length b and whose outer two have length at least two.

Temporarily suppose that no chain has length one. Away from the ends of the path, two chains of maximum length cannot then be adjacent. The largest average allowed by this restriction comes from alternating b and $b-1$. For example, at $n = 12$ the candidate sequence is $7, 6, 7, 6, \ldots$, with mean $13/2$. The old parameter is accordingly $t_0 = 2b - 1 = 2p_0 - 3$.

As before, this efficient repetition returns to an occupied class. The additional issue is how to count departures from it. A chain of length one can allow a neighboring pair of maximum lengths that would otherwise be forbidden. Such a short chain loses many units of length, and some of that loss must pay for the exception it creates. Counting only the number of nonalternating positions would discard this information.

5.1 The return of an alternating pair

The normalization of one pair is $h_{b-1}h_b$. It has an especially simple description after changing coordinates: advance seven places backwards around a circle, then interchange two neighboring symbols. A transposition either splits one cycle into two or joins two cycles into one. This gives the order without following a long word of cells.

Define

$$R_0 = \begin{cases} 2n/7, & 7 \mid n, \\ u(n-u), & 7 \nmid n, \end{cases} \tag{7}$$

where in the second case $1 \le u < n$ and $7u \equiv 1 \pmod{n}$. For $n = 12$, the inverse is $u = 7$, giving $R_0 = 35$.

Lemma 5.1. *An alternating factor of $2R_0 + 1$ chains, with lengths $b, b-1$ in either order, cannot occur in a genuine light segment. Moreover, $R_0 \le n^2/4$.*

Proof. Let s be the successor permutation of the circle $(1, 3, \dots, n, 2)$, and put $\tau = (1\ 3)$. The permutations in (3) give $\rho = \tau s$ and $g = \tau s^3$. Since g has order p_0,

$$M_0 = h_{b-1}h_b = \rho g^{-2} \rho g^{-1} = \tau s^{-2} \tau s^{-5} \tau.$$

Conjugation by $s^2\tau$ gives τs^{-7}, as described above. If $7 \nmid n$, the map s^{-7} is an n-cycle. The transposition cuts it into arcs of lengths u and $n-u$. These lengths are coprime, so the order is $u(n-u) \le n^2/4$. If $7 \mid n$, the map s^{-7} has seven cycles of length $n/7$. The transposition joins two and leaves the other five, giving order $2n/7 \le n^2/4$.

Reversing the pair gives a conjugate map with the same order. In an alternating factor of $2R_0+1$ chains, the first $2R_0$ have outgoing shortened gaps. Their R_0 pairs have combined normalization equal to the identity, so the following short-run start repeats the first start class. The post-B forcing rule applies because each such run has $n - 2 \ge 7$ blocks. □

5.2 Paying for the breaks in alternation

We first describe the count. Measure each chain's loss from its maximum length b. The two efficient lengths $b, b-1$ then become the binary symbols $0, 1$, and the efficient pattern becomes an alternating binary string. We shall remove larger losses, cut at adjacent equal binary symbols, and apply Lemma 5.1 to the alternating strings that remain.

In the interior of the path there are two kinds of equal adjacency. A pair 11 already contains extra loss: both chains have fallen below the maximum. A pair 00 has no such loss, but the four-chain restriction requires a neighboring chain of length one. That chain supplies the payment. The following proof keeps track of these two sources separately.

Proposition 5.2. *For $n = 3m$, $m \ge 3$, we have*

$$\mathrm{CP}\left(p_0 + \frac{1}{2} + \frac{1}{4R_0},\, t_0 - \frac{1}{2R_0}\right).$$

Proof. Consider a path of $q \ge 2$ chains. Write $z_i = b - \ell_i$ for the loss of its ith chain, and let w_i indicate whether that chain has length one. For each four-chain window, at least one of the middle chains is below maximum length or one of the outer chains has length one. Thus, for $2 \le i \le q-2$,

$$f_i := \mathbf{1}_{z_i > 0} + \mathbf{1}_{z_{i+1} > 0} + w_{i-1} + w_{i+2} \ge 1.$$

Think of each summand as a witness to this local restriction. An ordinary nonmaximum chain supplies one marker, and a chain of length one supplies two: it can witness the restriction in a middle position or in an outer position. Accordingly, put $v_i = \mathbf{1}_{z_i>0} + w_i$. Each marker occurs in at most two of the windows counted by $\sum f_i$. Also $v_i \le z_i$: a chain of length one loses $b-1 \ge 4$ units, more than enough to supply its two markers.

There are $q-3$ windows when $q \ge 3$. This suggests measuring the remaining loss by $E = 2\sum_i z_i - (q-3)$, after subtracting their mandatory witnesses. Let $A = \sum_{i=2}^{q-2}(f_i - 1)$ count the extra witnesses. Then

$$E = 2\sum_i (z_i - v_i) + \left(2\sum_i v_i - \sum_{i=2}^{q-2} f_i\right) + A.$$

All three terms are nonnegative. A nonmaximum chain other than a chain of length one spends one loss unit on its marker. If its loss is at least two, at least one unit remains. A chain of length one spends two of its $b-1$ units, leaving $b-3$. Consequently, if B_0 counts the indices with $z_i \ge 2$ and $\ell_i \ne 1$, and W counts those with $\ell_i = 1$, then

$$E \ge 2B_0 + 2(b-3)W + A. \tag{8}$$

This is the reserve available to pay for cuts.

Now restrict to the interior indices $2, \dots, q-1$ and remove those counted by B_0 or W, leaving breaks where they stood. The surviving strings are binary. Cut at every 00 or 11 adjacency. Each 11 contributes at least one extra witness to A. Each 00 needs a neighboring chain of length one, and a single such chain can explain at most two 00 adjacencies. Hence the total number of alternating strings is at most $B_0 + 3W + A + 1$: a removed chain creates one break, and a chain of length one may account for two further cuts.

Every remaining string has length at most $2R_0$. Counting the removed indices as well gives

$$q - 2 \le B_0 + W + 2R_0(B_0 + 3W + A + 1) \le 2R_0(E+1). \tag{9}$$

For the last inequality, (8) pays for each type of break separately; $b \ge 5$ and $R_0 \ge 1$ suffice for the coefficients. At $q = 3$ the window sums are empty and the argument still applies. At $q = 2$, the identity $E = 2\sum_i z_i + 1 \ge 1$ gives (9) directly.

Finally let $\mu = t_0/2 = b - 1/2$. The total length is

$$\sum_i \ell_i = \mu q + \frac{3}{2} - \frac{E}{2}. \tag{10}$$

Inserting (9) into (10) gives

$$\sum_i \ell_i \le \left(\mu - \frac{1}{4R_0}\right) q + 2 + \frac{1}{2R_0} = p_0 + \frac{1}{2} + \frac{1}{4R_0} + \frac{1}{2}\left(t_0 - \frac{1}{2R_0}\right)(q-1).$$

For a single chain, the earlier bound $\ell_1 \le p_0$ suffices. □

The average has decreased, but the endpoint allowance has increased slightly. The count explains both changes: alternating pieces have bounded length, while the two ends can leave incomplete pieces. Only the first cost accumulates with the number of chains. The same proof permits any real upper bound $U \ge R_0$ in place of R_0, because every alternating string has length at most $2R_0 \le 2U$. In particular, the simpler choice $U = n^2/4$ is valid.

6 The resulting lower bound

We have obtained the chain estimates needed by the deficit count. We now apply the earlier word reduction and give the parameters together. This is the same capacity calculation illustrated in (1), with the forest theorem accounting for the components and their endpoints.

For $p \geq 0$ and $t > 0$, define

$$\mathcal{B}_K(p,t) = F_3(K) + K - 3 + \left\lceil \frac{2(K-2)! - 2(K-2) - 4p/t}{K^2 - 4K + 1 - 2(K-1)/t} \right\rceil. \tag{11}$$

In each application below the fraction is positive. The preceding sections have supplied p and t by bounding the total length of a path of chains. The following proposition states precisely how that estimate enters the word bound.

Proposition 6.1 (Chain-to-word transfer)**.** *Let $n = K - 1 \geq 7$. Suppose p, t are real numbers satisfying*

$$0 \leq p \leq t, \qquad t > 0, \qquad t(n-4) \geq 4p + 6.$$

If $\mathrm{CP}(p,t)$ holds, then $S(K) \geq \mathcal{B}_K(p,t)$.

The proof combines the scalar deficit estimate with the hereditary forest theorem. For reference, it is included with the scalar argument in Appendices C and D. Its quantifier over all genuine segments is what permits the estimate to be used on the paths produced by the reduction from an arbitrary word.

The two return orders in the next theorem are R_0 from (7) and $R_1 = \mathrm{ord}(M_1)$ from (4). Their exact values can make the bound stronger at particular sizes. Replacing them by their common upper bound $n^2/4$ gives a simpler formula consisting of rational functions on the three residue classes.

Theorem 6.2. *For every $K \geq 8$, we have $S(K) \geq \mathcal{B}_K(p,t)$ with the following parameters. If $K \geq 13$, write $n = K - 1 = 3m + r$, where $0 \leq r \leq 2$, and use*

r	p	t
0	$2m + \frac{1}{2} + \frac{1}{4R_0}$	$4m - 3 - \frac{1}{2R_0}$
1	$m(2m+1)$	$2m^2 + 2m - 2 - \frac{1}{R_1}$
2	$(m+1)(2m+1)$	$4m^2 + 5m - \frac{5}{2}$

For $8 \leq K \leq 12$, use

K	8	9	10	11	12
p	10	15	521/80	21	28
t	46/3	287/12	359/40	263/12	195/4.

In the general table, either return order may instead be replaced by $n^2/4$.

Proof of Theorem 6.2. The chain inequalities are (5), (6), and Proposition 5.2. At $K = 8$, the earlier theorem gives $\mathrm{CP}(10, 10)$, which implies $\mathrm{CP}(10, 46/3)$. The numerical hypotheses of Proposition 6.1 are verified in Appendix E, including all five initial sizes. That proposition gives the word bound. The counts remain valid when a return order is replaced by any larger real number, proving the final assertion as well. □

Write $B(K)$ for the exact-order formula in Theorem 6.2 and $Q(K)$ for the quadratic-order formula. For $8 \le K \le 12$ these use the same initial rows. Every occurrence of a difference between these formulas below is an ordinary difference of integers.

6.1 The contribution of the return orders

To isolate the return-order improvements, define $O(K)$ by the same initial rows for $K < 13$ and, for $K \ge 13$, by $O(K) = \mathcal{B}_K(p, t)$ with the parameters

$n = K - 1$	p	t
$3m$	$2m$	$4m - 3$
$3m + 1$	$m(2m + 1)$	$2m^2 + 2m - 2$
$3m + 2$	$(m + 1)(2m + 1)$	$4m^2 + 5m - 5/2.$

Thus O already includes the collision between two efficient pairs in Section 4. The remaining differences come from the orders R_0 and R_1.

Proposition 6.3. *For every $K \ge 13$ with $K \equiv 1, 2 \pmod 3$, we have $B(K) > O(K)$. Along these two residue classes, respectively,*

$$\liminf_{K\to\infty} \frac{B(K) - O(K)}{(K-9)!} \ge \frac{9}{2}, \qquad \liminf_{K\to\infty} \frac{B(K) - O(K)}{(K-11)!} \ge 324.$$

Replacing B by Q, both normalized differences converge to the displayed constants.

Proof. Before taking ceilings, the exact-order improvements in the two cases are asymptotic to

$$\frac{9n!}{8R_0n^6}, \qquad \frac{81n!}{R_1n^8},$$

respectively. Appendix E gives the exact subtractions and verifies that the real improvement exceeds one from $n = 12$ and $n = 13$, respectively. The bound $R_i \le n^2/4$ gives the two lower limits, with equality in the limit for the quadratic choices. Taking ceilings changes each difference by less than one. □

The exact orders sometimes give larger gains: for instance, $R_0 = 2n/7$ when $7 \mid n$, and $R_1 = n$ in two residue classes of m. This is why Proposition 6.3 states lower limits for the strongest formula. At $K = 13, 14$, its gains are 12 and 58; at $K = 16, 17$, they are 9 528 and 16 920.

6.2 Comparison with the all-size reference bound

Halfar [2] proves the following all-size formula, which he attributes to Fritsch:

$$H(K) = F_3(K) + K - 3 + \left\lceil \frac{2((K-2)! - (K-2))}{K^2 - 4K + 1} \right\rceil, \qquad K \ge 7. \tag{12}$$

This is Theorem B of the proof package announced on 6 October 2026 [3]. His Theorem C gives stronger bounds from searched parameters at individual sizes. The comparison below is with the displayed formula H, not those finite bounds.

The earlier chain theorem already improves the all-size formula (12) at a larger order than the refinements above. We record the comparison to distinguish these two contributions.

Corollary 6.4. *For either $A = B$ or $A = Q$, we have*

$$A(K) = F_3(K) + (2 + o(1))(K - 4)!,$$

and, as $K \to \infty$ through the indicated residue class,

$$A(K) - H(K) = \begin{cases} (3 + o(1))(K - 6)!, & K \equiv 1 \pmod 3, \\ (18 + o(1))(K - 7)!, & K \equiv 2 \pmod 3, \\ (9 + o(1))(K - 7)!, & K \equiv 0 \pmod 3. \end{cases}$$

Proof. For both parameter choices, p/t is bounded and $t \geq 1$ for all sufficiently large K. The numerator inside the ceiling in (11) is therefore $2(K-2)! + O(K)$ and its denominator is $K^2(1 + O(1/K))$. Since $(K-2)!/K^2 \sim (K-4)!$, the claimed leading term follows. The additive term $K-3$ and the rounding error are $o((K-4)!)$. With $\kappa = n^2 - 2n - 2$, the difference of the real expressions inside the ceilings in (11) and (12) is exactly

$$\frac{4[n! - n(n-1) - p\kappa]}{t\kappa(\kappa - 2n/t)}.$$

In the three cases $n \equiv 0, 1, 2 \pmod 3$, the parameter t is asymptotic to $4n/3$, $2n^2/9$, and $4n^2/9$, respectively. Substitution gives the three constants. Rounding changes each difference by less than one. □

Corollary 6.5. *For every real $\varepsilon > 0$, there is K_0 such that every superpermutation on $K \geq K_0$ symbols has length at least*

$$F_3(K) + (2 - \varepsilon)(K - 4)!.$$

Proof. The normalized correction of $B(K)$ tends to 2, and every covering word has length at least $B(K)$ by Theorem 6.2. □

The conclusion concerns a lower bound, not an asymptotic formula for $S(K)$. In particular, the present argument does not produce a positive coefficient of $(K-3)!$. It strengthens the count of the cheapest local patterns while retaining the same segment-to-word reduction.

A The return order when $n = 3m + 1$

We calculate the order of the map M_1 in (4) for $m \geq 3$.

m mod 6	Cycle lengths defining R_1
$0, 2$	$3m + 1$
1	$(2m + 1)/3,\ (7m + 2)/3$
3	$(7m + 3)/6,\ (11m + 3)/6$
4	$(2m + 1)/3,\ (7m + 2)/6,\ (7m + 2)/6$
5	$(7m + 1)/6,\ (11m + 5)/6$

Table 1: The nonzero cycles of the return map when $n = 3m + 1$. The order R_1 is their least common multiple.

The nonzero cycles of g are

$$(3, 6, \ldots, 3m), \qquad (1, 5, 8, \ldots, 3m-1, 2, 4, 7, \ldots, 3m+1).$$

The first has length m and the second length $2m+1$. To evaluate g^{m-1} on the second, index its letters by

$$B_0 = 1, \quad B_j = 3j+2 \ (1 \le j < m), \quad B_m = 2, \quad B_{m+j} = 3j+1 \ (1 \le j \le m).$$

The map g adds one to these indices modulo $2m+1$. Also

$$(\rho g^{-1})(1) = 2, \quad (\rho g^{-1})(2) = 3m, \quad (\rho g^{-1})(3) = 3m+1, \quad (\rho g^{-1})(4) = 3, \quad (\rho g^{-1})(5) = 1,$$

and $(\rho g^{-1})(x) = x-2$ for $6 \le x \le 3m+1$. Applying g^{m-1} and then ρ gives

x	1	2	3	4	5	6	7	8	9
$M_1(x)$	$3m-1$	$3m-2$	$3m-3$	$3m+1$	$3m$	2	3	4	1,

and, for $10 \le x \le 3m+1$,

$$M_1(x) = \begin{cases} x-6, & x \equiv 0 \pmod 3, \\ x-5, & x \equiv 1 \pmod 3, \\ x-4, & x \equiv 2 \pmod 3. \end{cases} \tag{13}$$

For example, when $x = 3i \ge 6$, its image under ρg^{-1} is B_{m+i-1}. Adding $m-1$ to this index gives $2m$ at $i = 2$, zero at $i = 3$, and $i-3$ for $i \ge 4$. These are the exceptional values $M_1(6) = 2$, $M_1(9) = 1$, and the first line of (13). The other two residues follow by the same index substitution.

Every orbit outside $\{1, 2, 3, 4, 5\}$ decreases until it reaches that set. Thus every cycle contains one of these five *anchors*. The descent times are as follows; the first jump out of an anchor is not included.

Starting value	Anchor reached	Steps
$6j$ $(j \ge 1)$	2	j
$6j+3$ $(j \ge 1)$	1	j
$9j+1$ $(j \ge 1)$	5	$2j-1$
$9j+4$ $(j \ge 0)$	4	$2j$
$9j+7$ $(j \ge 0)$	3	$2j+1$
$9j+2$ $(j \ge 1)$	3	$2j$
$9j+5$ $(j \ge 0)$	5	$2j$
$9j+8$ $(j \ge 0)$	4	$2j+1$

For multiples of three, this is repeated subtraction of six, ending at $6 \mapsto 2$ or $9 \mapsto 1$. In the other residues, two steps subtract nine. The low values $4, 5, 7, 8, 10$ give the stated endpoints.

Write $m = 6h + r$. Applying the descent times to the five initial jumps gives the following first-return data, in anchor order $1, 2, 3, 4, 5$.

r	Destinations	Arc lengths
0	$(4, 3, 1, 5, 2)$	$(4h, 4h, 3h, 4h, 3h+1)$
1	$(3, 5, 2, 4, 1)$	$(4h+1, 4h, 3h+1, 4h+1, 3h+1)$
2	$(5, 4, 1, 3, 2)$	$(4h+1, 4h+1, 3h+1, 4h+2, 3h+2)$
3	$(4, 3, 2, 5, 1)$	$(4h+2, 4h+2, 3h+2, 4h+2, 3h+2)$
4	$(3, 5, 1, 4, 2)$	$(4h+3, 4h+2, 3h+2, 4h+3, 3h+3)$
5	$(5, 4, 2, 3, 1)$	$(4h+3, 4h+3, 3h+3, 4h+4, 3h+3)$.

Summing the arc lengths around each anchor cycle gives exactly Table 1. There are no further cycles, since every nonanchor reaches an anchor.

For a single cycle, the order is $n \le n^2/4$. For two cycles of lengths u, v, it is at most $uv \le n^2/4$. The three-cycle case has lengths u, v, v, so its order is at most $uv \le n^2/8$. This proves the quadratic bound used in the main theorem.

B The collision between two efficient pairs

This appendix proves Lemma 4.2. Retain the parameters $n = 3m + 2$, $p_0 = (m + 1)(2m + 1)$, $v = 2m(m + 1)$, $b = p_0 - 1$, and $c = v - 2$.

Proof. Normalize the first short-run entry to $(3, \dots, n, 2, 1, 0)$. In a canonical C cell, the nonzero cyclic core of its rth full run is

$$L_r = [\mathrm{rot}^{r-1}(1, 3, \dots, n-1, 2), n], \qquad 1 \le r \le n - 3.$$

Representatives beginning in 1 give

$$L_1 = [1, 3, \dots, n-1, 2, n], \qquad L_{n-3} = [1, 3, \dots, n-3, n, n-2, n-1, 2].$$

Set $M = h_c h_b$ and $j = v - m - 4$. Number cells from zero. The last full run of cell j in the second chain of the first pair has actual core $h_b^{-1} g^{-j} L_{n-3}$. The first full run of cell m in the second chain of the second pair has core $M^{-1} h_b^{-1} g^{-m} L_1$. Both cells precede their chains' final B cell, since $0 \le j \le c - 2$ and $m \le c - 2$.

Equality of these two cores is equivalent to $E L_{n-3} = L_1$, where

$$E = g^m h_b h_c g^{-j} = g^m \rho g^{-1} \rho g^{m+2}.$$

The last expression uses $g^{p_0} = 1$, $c = p_0 - m - 3$ and $j = p_0 - 2m - 5$. The two nonzero cycles of g are

$$(1, 5, 8, \dots, 3m + 2), \qquad (2, 4, 7, \dots, 3m + 1, 3, 6, \dots, 3m).$$

Applying the displayed expression for E gives

$$E(1) = 3, \quad E(2) = 1, \quad E(x) = x + 1 \ (3 \le x \le n - 3),$$
$$E(n-2) = 2, \quad E(n-1) = n, \quad E(n) = n - 1.$$

It sends L_{n-3} to $[3, 4, \dots, n-1, 2, n, 1]$, a rotation of L_1. Thus the two full runs visit the same two-loop, contradicting class distinctness. □

B.1 The initial case on nine symbols

At $m = 2$, the efficient tiles are (12), $(13, 11)$ and $(14, 10)$. Their transfers lie in a group of order 12. To see this directly, put $A = (2\ 8)$, $Q = (2\ 8\ 7)$, $U = (1\ 3)$ and $V = (5\ 6)$. The three transfers are AV, AU, QUV. They belong to

$$\{\sigma U^e V^f : \sigma \in \mathfrak{S}_{\{2,7,8\}},\ e, f \in \{0, 1\},\ \mathrm{sgn}(\sigma) = (-1)^{e+f}\},$$

a group of order 12. Twelve consecutive efficient interior tiles would give thirteen prefix maps in this group, two equal, and hence a repeated start class. Positive losses are integers and the final allowance is two. Lemma 4.1 therefore gives $\mathrm{CP}(15, 24 - 1/12) = \mathrm{CP}(15, 287/12)$.

C The scalar conversion

For completeness, we give the counting step that turns a chain inequality into a hereditary bound on segment deficits. This is Pantone's scalar theorem [1]; no integrality of p or t is required.

Call a finite sequence of nonnegative integers *legal* if its zero intervals have length at most a, except that an entirely zero sequence may have length $a+1$; every factor $(0,1,d)$ has $d \geq 2$; and across an internal zero interval of length a, the neighboring positive entries sum to at least $a+2$. These are the local run rules for genuine light segments. A bound on all legal sequences satisfying $\mathrm{CP}(p,t)$ therefore applies to those segments.

Lemma C.1. *Let $a \geq 4$, $0 \leq p \leq t$, $t > 0$, and $t(a-2) \geq 4p+6$. Set*

$$\lambda = at - 2, \qquad B = 2t(a+1) + 4p.$$

Every legal sequence satisfying $\mathrm{CP}(p,t)$ on its zero-ended factors has length r and sum D satisfying $\lambda D - 2tr + B \geq 0$.

Proof. First consider a zero-ended sequence with $s \geq 1$ positive clusters. Let cluster j have length m_j and sum D_j. The rule concerning $(0,1,d)$ gives $D_j \geq m_j + 1$, so $D_j - 2 \geq m_j - 1 \geq 0$. Moreover $D_j = 2$ holds exactly for a regular cluster. Put

$$M = \sum_j (m_j - 1), \quad V = \sum_j (D_j - 2), \quad b = \#\{j : (m_j, D_j) \neq (1,2)\}.$$

Write $u_0, \ldots, u_s$ for the zero-gap lengths, including the two end gaps, and define

$$g_0 = \#\{j : 1 \leq j < s,\ u_j = a\}, \quad H = \sum_{j=1}^{s-1} \max(0, a-1-u_j), \quad L = (a-u_0) + (a-u_s).$$

Counting entries and deficit gives

$$r = as + a + 1 + M - H + g_0 - L, \qquad D = 2s + V. \tag{14}$$

Let h_1 count the gaps of length $a-2$ between originally adjacent regular clusters, and let h_2 count the shorter such gaps. Cutting at the b other clusters and at these h_2 gaps leaves at most $b+1+h_2$ paths of chains. The sum of their numbers of links is h_1. Thus $\mathrm{CP}(p,t)$ gives

$$s - b \leq p(b+1+h_2) + \frac{t}{2} h_1.$$

Since $H \geq h_1 + 2h_2$ and $p \leq t$, it follows that

$$J := 4p + 4(p+1)b + 2tH - 4s \geq 0. \tag{15}$$

Set $c = \lambda - 2t$. The numerical hypothesis says $c \geq 4(p+1)$. Give cluster j the reserve

$$\Phi_j = \lambda(D_j - 2) - 2t(m_j - 1) - 4(p+1)\mathbf{1}_{(m_j, D_j) \neq (1,2)}.$$

A regular cluster has reserve zero. Every other cluster has $D_j - 2 \geq 1$, and therefore $\Phi_j \geq c(D_j - 2) - 4(p+1) \geq 0$.

Suppose a zero gap of length a separates clusters i, j. The actual neighboring positive entries sum to at least $a+2$; each other entry in those clusters contributes at least one. Hence

$$(D_i - 2) + (D_j - 2) \geq a - 2 + (m_i - 1) + (m_j - 1).$$

It follows that

$$\Phi_i + \Phi_j \geq \lambda(a-2) - 8(p+1) \geq \lambda(a-2) - 2c = \lambda(a-4) + 4t \geq 4t.$$

Form a graph on the positive clusters, with an edge across each gap of length a. Its g_0 edges form a subgraph of a path. In particular, every degree is at most two. Substituting (14) and (15) gives the identity

$$\lambda D - 2tr + B = J + 2tL + \frac{1}{2} \sum_{\{i,j\} \in E} (\Phi_i + \Phi_j - 4t) + \frac{1}{2} \sum_j (2 - \deg(j)) \Phi_j.$$

Every term on the right is nonnegative.

An entirely zero sequence has length at most $a+1$, so its left side is at least $4p$. An entirely positive sequence satisfies $\lambda D - 2tr \geq 0$. In a mixed sequence, remove the initial and final positive clusters. Each removed entry contributes at least $\lambda - 2t = c \geq 0$. The remainder is either one zero interval or a zero-ended sequence of the kind already considered. The empty sequence also satisfies the claimed inequality. □

D The hereditary forest estimate

This appendix explains why the local deficit inequality applies to an arbitrary covering word. A shortest word gives a minimum overlap path, but that path need not visit every rotation class consecutively. The forest construction replaces it by auxiliary complete-block paths and records the cost of the replacement. The classes in different components partition all the rotation classes. A block borrowed at the entrance to one component is required to be new only to that component.

The construction uses a particular minimum path. Call a minimum path *reduced* if it has no edge from u to its second cyclic rotation $\sigma^2(u)$. Such a path exists. Among the minimum paths, choose one with as many weight-one edges as possible. If it contained $u \to \sigma^2(u)$, remove $\sigma(u)$ from its other location and insert it between these two vertices. Removal cannot increase the weight, by the triangle inequality, and insertion replaces one weight-two edge by two weight-one edges. Moreover, removal deletes no weight-one edge. The only possible weight-one predecessor of $\sigma(u)$ is u, whose successor is already $\sigma^2(u)$, and the only possible weight-one successor is $\sigma^2(u)$, whose predecessor is already u. The new path would therefore still be minimum and have more weight-one edges, a contradiction.

We state the properties of the forest construction for this reduced minimum path. The construction and its residual inequality are inherited from the lower-bound framework [1]. The summation argument after the statement is included here.

Lemma D.1 (Forest decomposition)**.** *Put $K = n+1 \geq 8$ and $T = n!$. A reduced minimum overlap path determines a finite set of complete-block path components with a distinguished root. Each nonroot component is classified as cheap or expensive. For component v, let b_v be its number of blocks, q_v its number of runs, D_v its total deficit, m_v its number of light segments, and d_v^-, d_v^+ its first and last deficits. There are nonnegative integers A_v and y_v such that*

$$\sum_v b_v = T, \qquad b_v = nq_v - D_v, \qquad z_v = q_v + A_v, \qquad A_v \geq m_v - 1.$$

For the root and cheap components, $y_v = z_v$; for expensive components, $y_v \geq z_v + 1$.

For a cheap component with at least two runs, one can prepend an entrance block to obtain another class-distinct path. Its block count increases by one, its number of runs and z_v are unchanged,

and its first deficit decreases by one. Its last deficit is unchanged. A cheap component with one run has $y_v = 1$ *and* $d_v^- = d_v^+ \geq 1$.

There is a directed absorption relation on the components. Each component has at most one incoming and one outgoing edge, and every target is a cheap nonroot component. Every edge is either an isolated loop at a cheap one-run component of deficit one, or an ordinary edge $v \to w$ *with*

$$d_v^+ \geq 2, \qquad d_w^- \geq 2, \qquad d_v^+ + d_w^- \geq n + 1.$$

If V *is the number of cheap nonroot components with no incoming edge, and* $R = \sum_v y_v$, *then*

$$S(K) \geq K + (K+1)T - 3 + R + V. \tag{16}$$

The last inequality accounts for the original path as well as the auxiliary components. In particular, it does not assume that entrance blocks borrowed by different components are distinct. This is the reason for proving the local inequality on every genuine segment, including consecutive factors of an entrance-extended path.

Theorem D.2 (Hereditary forest estimate)**.** *Suppose* $0 < \delta < 1$ *and* $1 \leq \beta \leq \alpha := n - \delta$. *If every genuine light segment with* r *runs and total deficit* D *satisfies* $D \geq \delta r - \beta$, *then*

$$S(K) \geq K + (K+1)n! - 3 + \left\lceil \frac{n! - \beta}{n - \delta} \right\rceil.$$

Proof. The first step recovers endpoint information from the hypothesis. Write $F(d) = \max(0, d - \delta)$. Deleting the first deficit from a segment and applying the hypothesis to the remainder gives an extra term $d^- - \delta$. Applying the hypothesis without deletion gives an extra term zero. Taking the larger of these two estimates gives

$$D \geq \delta r - \beta + F(d^-).$$

There is an identical estimate at the last endpoint. When $r \geq 2$, we can delete either endpoint, both, or neither, and obtain

$$D \geq \delta r - \beta + F(d^-) + F(d^+). \tag{17}$$

The empty remainder causes no difficulty because $0 \geq -\beta$. The two endpoints in (17) must be distinct; we shall handle one-run components separately.

Apply Lemma D.1 and put $s_v = b_v - \alpha y_v$. Summing the segment estimates within one component gives

$$b_v - \alpha z_v \leq m_v \beta - \alpha A_v - F(d_v^+) \leq \beta - F(d_v^+),$$

because $A_v \geq m_v - 1$ and $\alpha \geq \beta$. Thus

$$s_v \leq \beta - F(d_v^+) \quad \text{at the root,} \qquad s_v \leq \beta - \alpha - F(d_v^+) \quad \text{at an expensive component.}$$

For a cheap component with at least two runs, apply the same argument to its entrance-extended path and use both endpoint estimates. Define $A(d) = \max(0, d - 1 - \delta)$ and $c = \beta - 1$. The additional block gives

$$s_v \leq c - A(d_v^-) - F(d_v^+).$$

If the two ends lie in different light segments, apply the one-end estimate to each of those segments. If they lie in the same segment, that segment has at least two runs and (17) applies. For a cheap component with a single run, the exact identity is

$$s_v = (n - d_v^-) - (n - \delta) = \delta - d_v^- = -F(d_v^-).$$

Put $\widehat{A}(d) = \min(A(d), c)$. For a cheap component, let $i_v, o_v \in \{0, 1\}$ indicate whether it has an incoming and an outgoing absorption edge. Then both the multi-run and single-run estimates imply

$$s_v \le c - i_v \widehat{A}(d_v^-) - o_v F(d_v^+).$$

For a single run, this follows from $0 \le \widehat{A} \le c$. Clipping the incoming charge is what permits one formula to cover that case without counting its only deficit twice.

Assign an intercept c_v to each component: use β at the root, c at cheap nonroots, and $\beta - \alpha$ at expensive nonroots. Assign the incoming charge $a_v = \widehat{A}(d_v^-)$ at cheap vertices and zero elsewhere, and the outgoing charge $b'_v = F(d_v^+)$. There is one exception. At a vertex carrying an isolated special loop, use $c_v = \delta - 1$ and $a_v = b'_v = 0$; its surplus is exactly $\delta - 1$.

The preceding bounds give

$$s_v + i_v a_v + o_v b'_v \le c_v.$$

For an ordinary edge $v \to w$, its two deficits give

$$A(d_w^-) + F(d_v^+) \ge n - 2\delta, \qquad F(d_v^+) \ge 2 - \delta.$$

Consequently its charges cover the target intercept, since

$$b'_v + a_w - c_w = \min\{F(d_v^+) + A(d_w^-) - c,\ F(d_v^+)\} \ge \min\{n + 1 - \beta - 2\delta,\ 2 - \delta\} > 0.$$

For a special loop, the corresponding excess is $1 - \delta > 0$.

Now sum the vertex inequalities and the edge inequalities. Each incoming and outgoing charge occurs exactly once on either side and cancels. The remaining intercepts belong to vertices with no incoming edge. These are the root, the V unabsorbed cheap vertices, and all the expensive vertices. The latter have nonpositive intercept. It follows that

$$T - \alpha R = \sum_v s_v \le \beta + (\beta - 1)V.$$

Since $\beta \le \alpha$, this implies $T \le \alpha(R + V) + \beta$. The number $R + V$ is an integer, so (16) gives the asserted ceiling bound. □

D.1 From the chain estimate to word length

Proof of Proposition 6.1. Set

$$\lambda = (n - 2)t - 2, \qquad \delta = \frac{2t}{\lambda}, \qquad \beta = \frac{2t(n - 1) + 4p}{\lambda}.$$

Lemma C.1 gives $D \ge \delta r - \beta$ for the total deficit D of every segment with r runs. The same estimate holds on its consecutive run factors. The parameter assumptions give $0 < \delta < 1$ and $1 < \beta < n - \delta$. For the latter inequality, use $4p \le t(n - 4) - 6$ and $t(n - 4) \ge 6$ to obtain

$$n(n - 4)t - 4p - 2n \ge (n - 1)(n - 4)t + 6 - 2n \ge 4n > 0.$$

Theorem D.2 therefore gives

$$S(K) \ge K + (K + 1)n! - 3 + \left\lceil \frac{n! - \beta}{n - \delta} \right\rceil. \tag{18}$$

Substitution and removal of the integer $(n - 1)!$ from the ceiling give (11). □

E Parameter checks and integer improvements

E.1 The transfer hypotheses

The chain inequalities are (5), (6), and Proposition 5.2. We verify the numerical assumptions of Proposition 6.1.

For $n = 3m \geq 9$, write $p_0 = 2m$. Since $R_0 \geq 1$, the new parameters satisfy

$$t - p = p_0 - \frac{7}{2} - \frac{3}{4R_0} > 0,$$

and

$$t(n-4) - 4p - 6 \geq (2p_0 - 3)(n-4) - 4p_0 - 8 - \frac{n-2}{2} = \frac{4}{3}n^2 - \frac{23}{2}n + 5 > 0.$$

The last polynomial is positive and increasing for $n \geq 9$.

For $n = 3m + 1$ with $m \geq 3$, the new t is at least $t_0 - 1 \geq p_0$, and

$$(t_0 - 1)(n-4) - 4p_0 - 6 = 6m^3 - 8m^2 - 19m + 3 > 0.$$

Writing $m = 3 + x$ makes the right side $6x^3 + 46x^2 + 95x + 36$. For $n = 3m + 2$, $m \geq 2$, all the stated choices have $t \geq t_0 - 1$. Here

$$t_0 - 1 - p_0 = 2(m+2)(m-1) > 0,$$

and

$$(t_0 - 1)(n-4) - 4p_0 - 6 = (m-2)(12m^2 + 23m + 15) + 26 > 0.$$

At $K = 8$, the earlier chain theorem gives $\mathrm{CP}(10, 10)$ [1]; increasing t to $46/3$ preserves this inequality and gives $t(n-4) = 4p + 6$. The values for $K = 9, 12$ are the two initial cases in Section 4. Those for $K = 10, 11$ are the first instances of the other two residue formulas.

Proposition 6.1 now proves the asserted word bound. Replacing either return order by any larger real number preserves the counts in Lemma 4.1 or (9). The weaker quadratic choices still satisfy the same numerical estimates, so they too give the claimed bounds.

E.2 Exact changes in the bound

Put $T = n!$ and $\kappa = n(n-2) - 2$. The nonconstant real term in (18) is

$$f(p, t) = \frac{t((n-2)T - 2(n-1)) - 2T - 4p}{\kappa t - 2n}.$$

First suppose $n = 3m + 1$. With $h = 1/R_1$, direct subtraction gives

$$f(p_0, t_0 - h) - f(p_0, t_0) = \frac{4h[T - p_0\kappa - n(n-1)]}{(\kappa(t_0 - h) - 2n)(\kappa t_0 - 2n)}. \tag{19}$$

For $n \geq 10$, we have $t_0 < n^2/4$, $p_0 < n^2$, $\kappa < n^2$, and $h \geq 4/n^2$. Hence this difference exceeds

$$256\frac{n! - n^4 - n^2}{n^{10}}.$$

At $n = 13$ the expression exceeds one, and it increases thereafter: $n!/n^{10}$ increases since $(1+1/n)^9 < 3 < n$, while n^{-6} and n^{-8} decrease. Thus the integer ceiling improves at every stated size in this

residue class. Since $t_0 \sim 2n^2/9$ and $\kappa \sim n^2$, (19) is asymptotic to $81n!/(R_1 n^8)$. The quadratic upper bound on R_1 gives the constant 324.

For $n = 3m$, set $h = 1/(2R_0)$, $d = 1/2 + h/2$, and $D_0 = \kappa t_0 - 2n$. Subtraction now gives

$$f(p_0 + d, t_0 - h) - f(p_0, t_0) = \frac{4h[T - n(n-1) - \kappa p_0 - (R_0 + 1/2)D_0]}{D_0(D_0 - \kappa h)}.$$

Using $R_0 \le n^2/4$, $p_0 = 2n/3$, $t_0 < 4n/3$ and $\kappa < n^2$, the total subtracted from T is less than n^5. The difference therefore exceeds

$$\frac{9}{2}\frac{n! - n^5}{n^8}.$$

This is greater than one at $n = 12$ and increases thereafter. Its factorial term increases because $(1+1/n)^7 < 3 < n$; the subtracted term decreases. More precisely, the exact difference is asymptotic to $9n!/(8R_0 n^6)$, giving the constant $9/2$. The difference of two ceilings changes an unrounded difference by less than one. Finally, $n!/n^j \sim (n-j)!$, and $K = n+1$, which gives both normalizations in the statement.

F The formal statements

The companion Lean development separates its public definitions from the proofs. The coverage predicate on a word w over $\mathrm{Fin}\, K$ is the assertion that every list p of length K with distinct entries has a prefix u and a suffix v with $w = upv$. Thus the final statements refer directly to covering words, with no path or chain hypothesis.

The two statements in `Submissions/General/Challenge.lean` are Theorem 6.2 with the exact and quadratic return orders. Their formulas include the initial rows $8 \le K \le 12$. Their declaration names are `exact_word_bound` and `quadratic_word_bound`, in the namespace `Superpermutation Results.GeneralLower`. The challenge imports only Mathlib. Its solution environment imports proved declarations with the same names; the internal bridge between the displayed formulas and the proof's formulas is checked by reflexivity. The Lean formula uses the natural-number ceiling. The correction in (11) is positive in every stated application, so this agrees with the ordinary ceiling used here.

The challenge in `Submissions/Asymptotics/Challenge.lean` states thirteen consequences. The correspondence with the paper is recorded below. Names in the second column are the suffixes of declarations in `SuperpermutationPalomar.LowerAsymptotics`.

Mathematical statement	Lean declaration
The leading correction in Corollary 6.4	`exact_leading_limit` `quadratic_leading_limit`
Corollary 6.5	`word_lower_bound_two_sub_epsilon`

The three comparisons of B with H	`exact_reference_limit_zero`
	`exact_reference_limit_one`
	`exact_reference_limit_two`

The three comparisons of Q with H	`quadratic_reference_limit_zero`
	`quadratic_reference_limit_one`
	`quadratic_reference_limit_two`

The two limits for $Q - O$	`quadratic_gain_zero_limit` `quadratic_gain_one_limit`
The two eventual lower estimates for $B - O$	`exact_gain_zero_asymptotic_lower` `exact_gain_one_asymptotic_lower`

Here the suffixes zero, one and two refer to $n = K - 1$ modulo three. The last two declarations say that, for each positive epsilon, the normalized difference is eventually at least the stated constant minus that epsilon. They do not assert convergence. The finite strict-improvement thresholds in Proposition 6.3 have the ordinary proof in Appendix E; they are not separate claims in this asymptotic challenge.

Every declaration just listed, and both general word bounds, has a proof using only `propext`, `Classical.choice`, and `Quot.sound`. No native computation assumption is used in these lower-bound families.

References

[1] J. Pantone, *The lower-bound framework for superpermutations*, companion Lean library, namespace `SuperpermutationLowerBound`. The paths below are relative to `proofs/SuperpermutationLowerBound/`. The geometric inputs are in `Refinement/ReturnGeometry.lean`, `Refinement/ReturnLocalCollisions.lean`, and the `Refinement/ChainPacking.lean` family of modules. The forest reduction is in the `Section01`–`Section10` directories and `Refinement/ForestResidual.lean`; the hereditary assembly is in `Refinement/ForestAssembly.lean`.

[2] J. Halfar, *Three lower bounds in Lean: Theorems A, B and C*, lower-bound proof source, revision `cc5c763bd5854317b20dadb138990802656e6f03`, October 2026.

[3] J. Halfar, *New upper bound with 1771/3456 and new lower bounds for $n = 8$ to 14, both in Lean*, announcement to the Superpermutators group, 6 October 2026. https://groups.google.com/g/superpermutators/c/6ozhk3faHDA.